

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Black-capped Chickadee"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

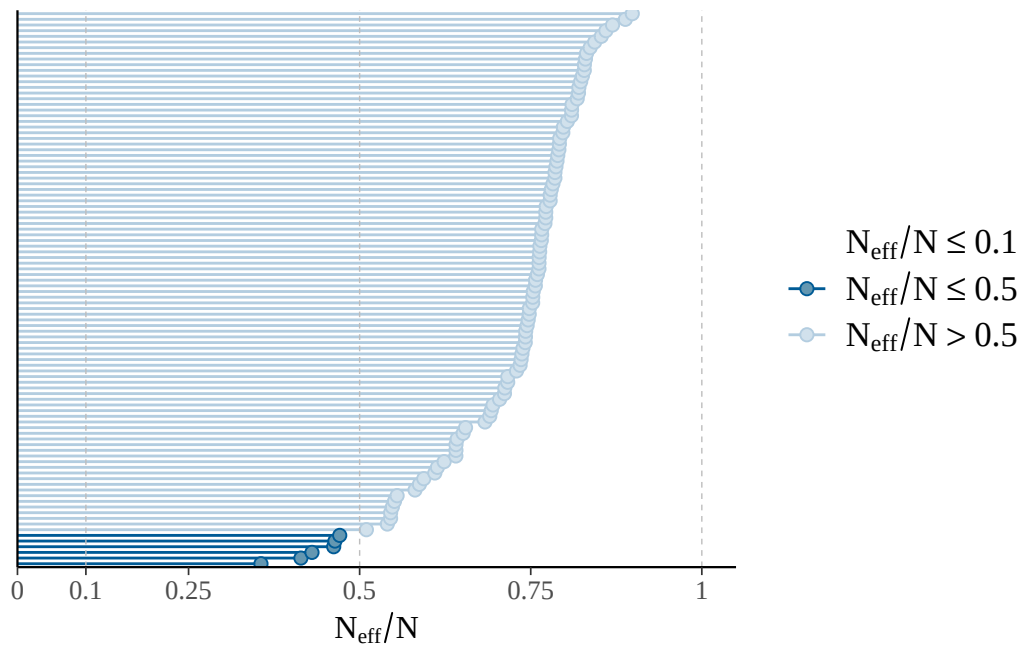
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4142	0.8560	1.1449	1.2033	1.4610	2.9030

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4540	0.7154	0.7621	0.7466	0.7955	0.8983

```
[1] "effective sample size in bulk and tail fine"
```

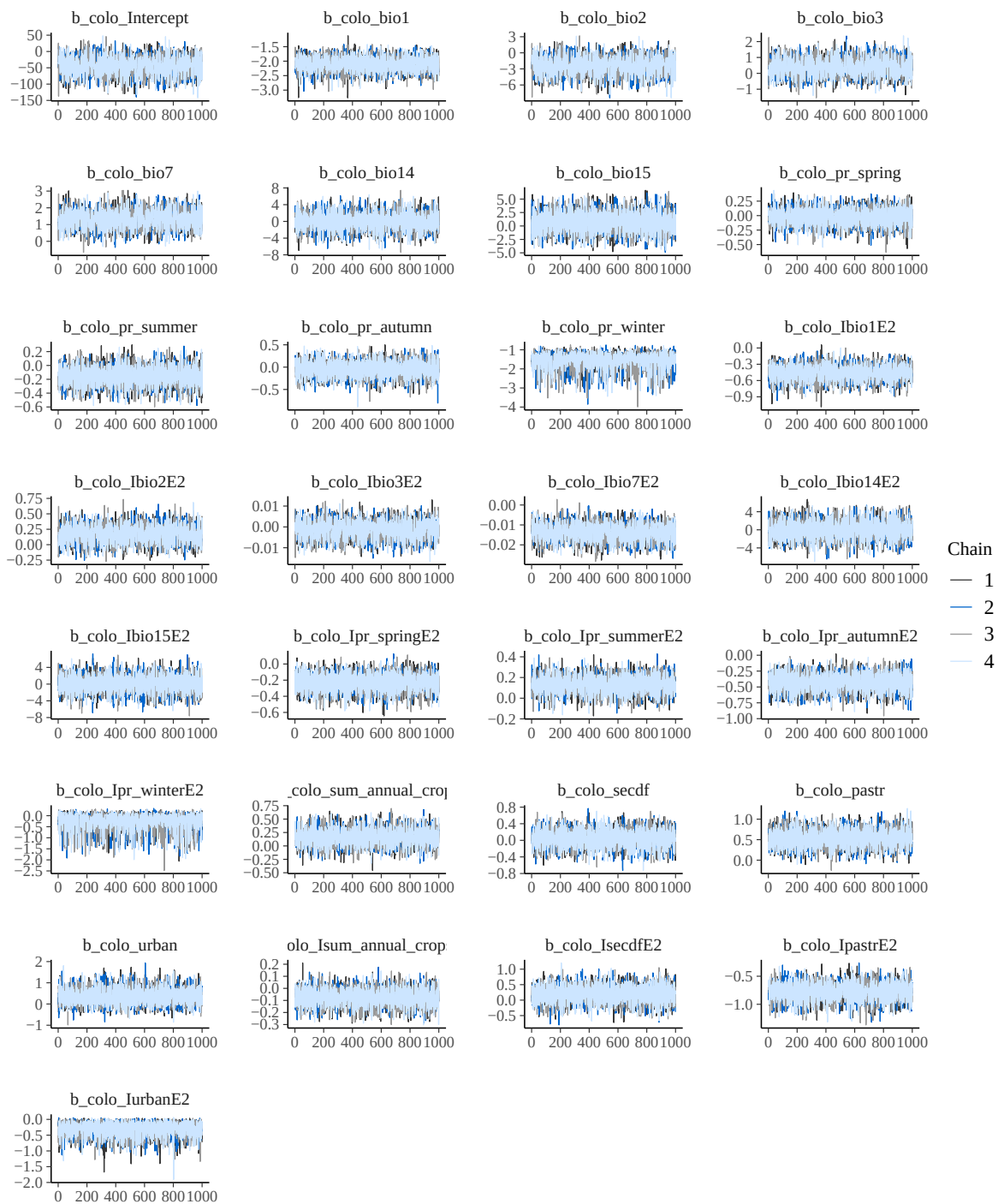
Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



Post-warmup iteration



Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9995	0.9999	1.0002	1.0005	1.0028

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio1_3yrs"      "b_occ_pr_spring_3yrs" "b_occ_pr_autumn_3yrs"
[4] "b_occ_pr_winter_3yrs" "b_occ_Ibio1_3yrsE2"   "b_occ_Ibio3_3yrsE2"
```

```
[1] "b_colo_bio1"          "b_colo_bio7"          "b_colo_pr_winter"
[4] "b_colo_Ibio1E2"       "b_colo_Ibio7E2"       "b_colo_Ipr_springE2"
[7] "b_colo_Ipr_autumnE2"  "b_colo_pastr"         "b_colo_IpastrE2"
[10] "b_colo_IurbanE2"
```

```
[1] "b_ex_pr_summer"      "b_ex_Ipr_springE2"    "b_ex_sum_annual_crops"
[4] "b_ex_secdf"
```

